

1.Project Title

Commodity Rate Management System.

2.Project Objective

The system allows users to view commodity by selecting State, District, Mandi, and Commodity.

3.Project Requirement

i.Functional -requirements

User-

- 1.user can select state***
- 2.user can select district***
- 3.user can select mandi***
- 4.user can select commodity***
- 5.user can view commodity rates***

Admin-

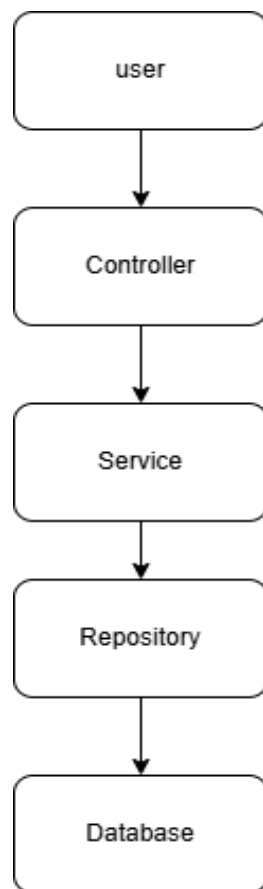
- 1.admin can add state***
- 2.admin can add district***
- 3.admin can add mandi***
- 4.admin can add commodity***
- 5.admin can update commodity price***

ii.Non-Functional Requirements

- 1. Simple ui (user interface).***

4. System design

Architecture



5. Use Cases

Actor:User-

-select state

-select district

-select mandi

-select commodity

Actor:Admin-

-add state

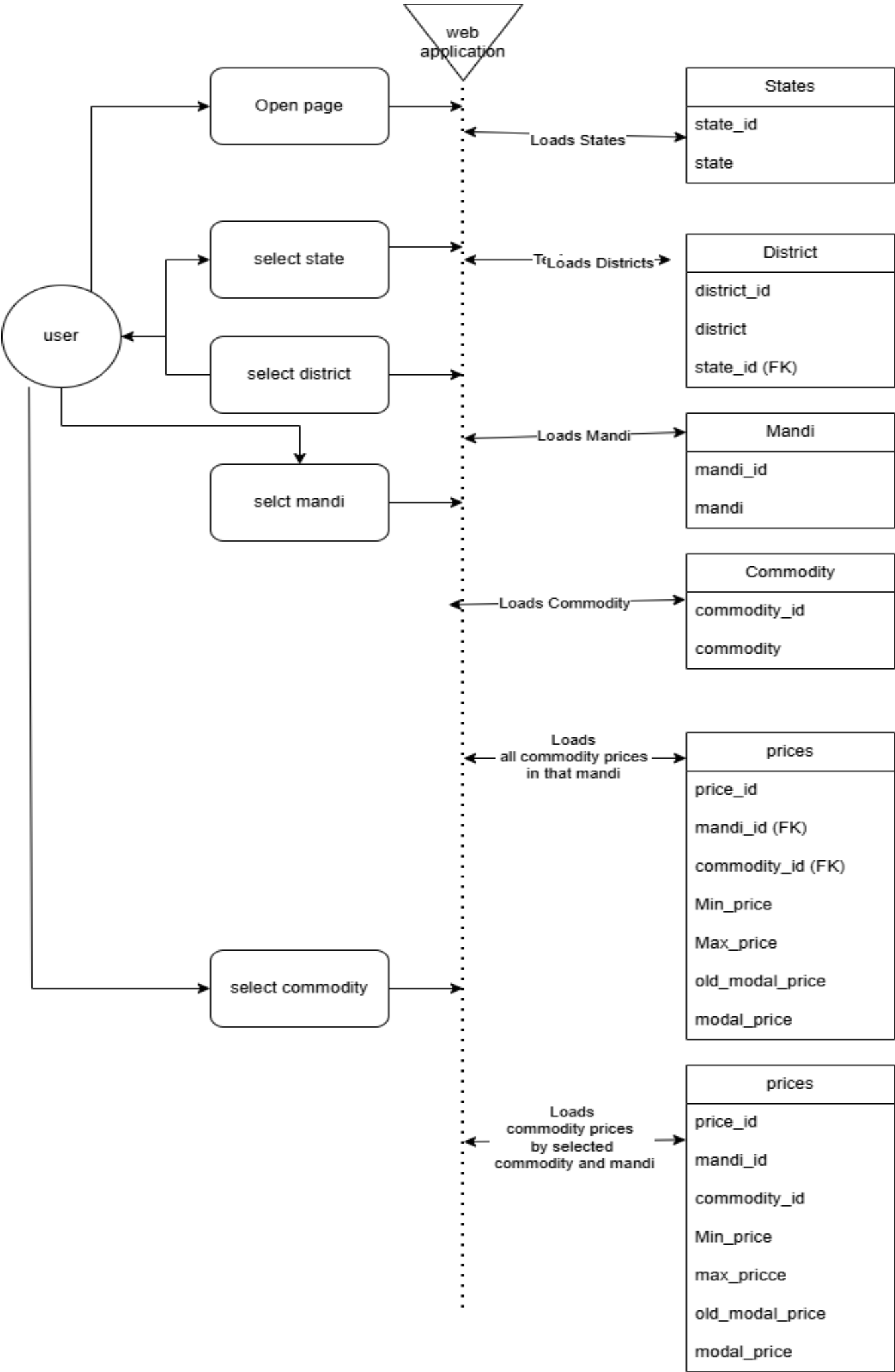
-add district

-add mandi

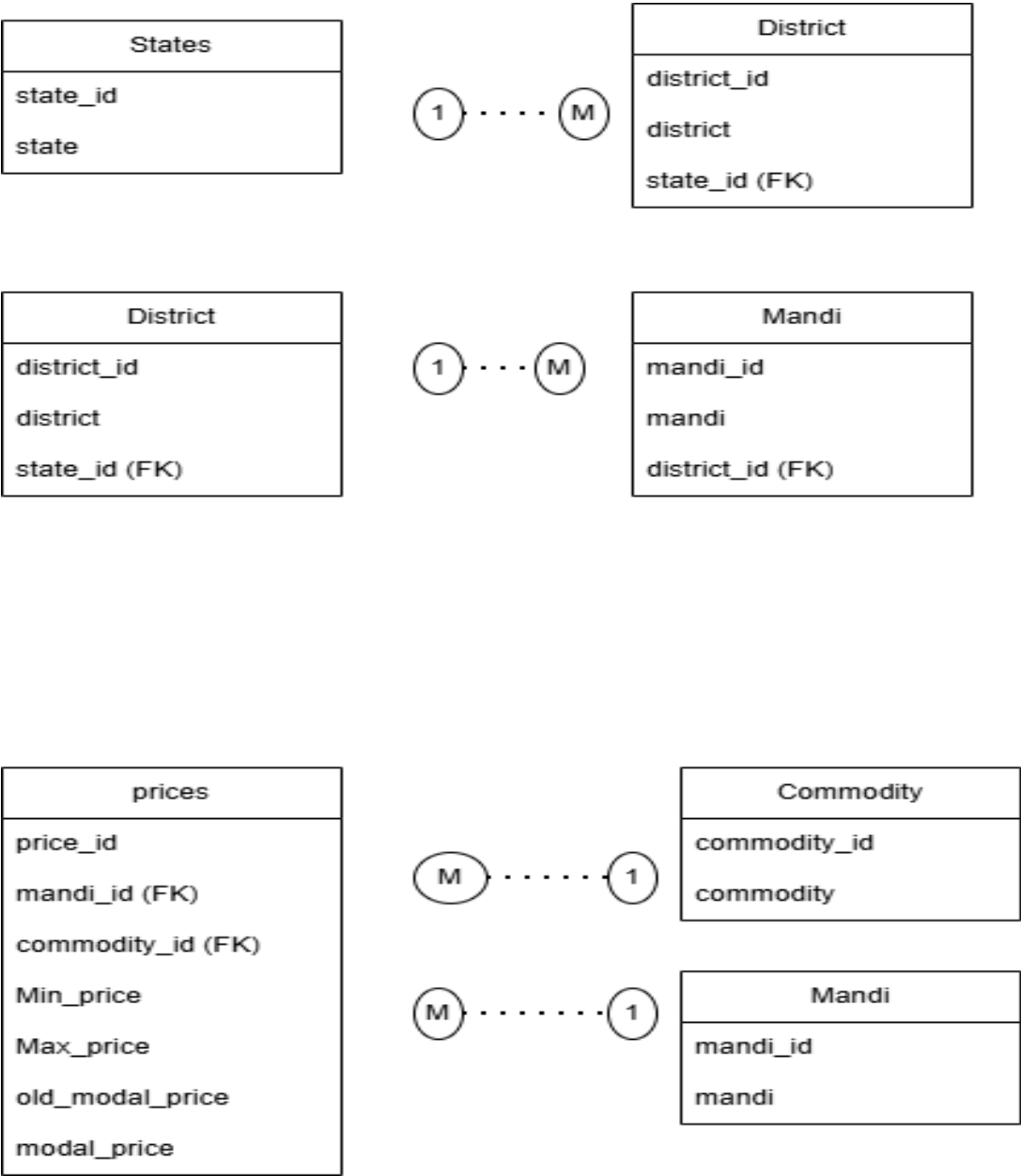
-add commodity

-update commodity price

6.Work Flow



7.Database design



API DESIGN

Method	API	Purpose
GET	/getStates	get States
GET	/getDistricts?id=state_id	Get districts in selected states
GET	/getMandi?id=distriCt_id	Get Mandis in selected mandi
GET	/getCommodity	Get Commodity
GET	/getPrice?commodity_id&mandi_id	Get commodity price that user selected
GET	/getPriceAll?mandi_id	Get all commodity price in mandi

Test Cases

User Test cases

TC ID	Scenario	Excepted Result
Tc_1	Open page	Load states
Tc_2	Select states	Load districts
Tc_3	Select district	Load mandi
Tc_4	Select mandi	Show all commodity price in that mandi and load Commodity
Tc_5	Select commodity	Show price of that commodity in that area.
Tc_6	Select district	Disable mandi
Tc_7	Select state	Disable mandi and district

Service Layer Test Cases

TC ID	Method	Scenario	Expected Result
TC-01	getStates()	States exist	Return list of states
TC-02	getStates()	No states available	Throw ResourceNotFoundException("states is empty")
TC-03	getStates()	Repository returns null	Throw ResourceNotFoundException("states is empty")
TC-04	getDistricts(id)	State exists and has districts	Return district list
TC-05	getDistricts(id)	State not found	Throw ResourceNotFoundException("are you looking for which state it is not found")
TC-06	getDistricts(id)	State found but district list empty	Throw ResourceNotFoundException("district is empty")
TC-07	getDistricts(id)	State found but district list null	Throw ResourceNotFoundException("district is empty")
TC-08	getMandi(id)	District exists with mandi list	Return mandi list
TC-09	getMandi(id)	District not found	Throw ResourceNotFoundException("are you looking for which district it is not found")
TC-10	getMandi(id)	Mandi list empty	Throw ResourceNotFoundException("mandi is empty")
TC-11	getMandi(id)	Mandi list null	Throw ResourceNotFoundException("mandi is empty")
TC-12	getCommodity()	Commodity exists	Return commodity list
TC-13	getCommodity()	Commodity list empty	Throw ResourceNotFoundException("commodity is empty")
TC-14	getCommodity()	Commodity list null	Throw ResourceNotFoundException("commodity is empty")
TC-15	getPrice(cid, mid)	Mandi, Commodity and Price exist	Return Price object
TC-16	getPrice(cid, mid)	Mandi not found	Throw ResourceNotFoundException("mandi is not found")
TC-17	getPrice(cid, mid)	Commodity not found	Throw ResourceNotFoundException("commodity is not found")
TC-18	getPrice(cid, mid)	Price not found	Throw ResourceNotFoundException("Price is not found")

TC-19	getPriceAll(mid)	Mandi and prices exist	Return price list
TC-20	getPriceAll(mid)	Mandi not found	Throw ResourceNotFoundException("mandi not found")
TC-21	getPriceAll(mid)	Price list empty	Throw ResourceNotFoundException("pricae not found")
TC-22	getPriceAll(mid)	Price list null	Throw ResourceNotFoundException("pricae not found")
TC-23	validateList()	Valid list	Return the same list
TC-24	validateList()	Empty list	Throw ResourceNotFoundException
TC-25	validateList()	Null list	Throw ResourceNotFoundException
TC-26	validateObject()	Valid object	Return the same object
TC-27	validateObject()	Null object	Throw ResourceNotFoundException

Controller Layer Test Cases

TC ID	Method/Endpoint	Scenario	Expected Result
CT-01	/mandiapp	User opens mandi application page	Return mandiapp view
CT-02	/getStates	States are available	Return list of states
CT-03	/getStates	No states available	Return 404 / ResourceNotFoundException
CT-04	/getDistricts?id	Valid state id	Return list of districts
CT-05	/getDistricts?id	Invalid state id	Return 404 / ResourceNotFoundException
CT-06	/getDistricts?id	State exists but districts are empty	Return 404 / ResourceNotFoundException
CT-07	/getMandi?id	Valid district id	Return list of mandis
CT-08	/getMandi?id	Invalid district id	Return 404 / ResourceNotFoundException
CT-09	/getMandi?id	District exists but mandi list is empty	Return 404 / ResourceNotFoundException
CT-10	/getCommodity	Commodities are available	Return commodity list
CT-11	/getCommodity	Commodity list is empty	Return 404 / ResourceNotFoundException
CT-12	/getPrice?cid&mid	Valid commodity id and mandi id	Return price object
CT-13	/getPrice?cid&mid	Invalid mandi id	Return 404 / ResourceNotFoundException
CT-14	/getPrice?cid&mid	Invalid commodity id	Return 404 / ResourceNotFoundException
CT-15	/getPrice?cid&mid	Price data not available	Return 404 / ResourceNotFoundException
CT-16	/getPriceAll?mid	Valid mandi id	Return list of prices
CT-17	/getPriceAll?mid	Invalid mandi id	Return 404 / ResourceNotFoundException
CT-18	/getPriceAll?mid	Price list is empty	Return 404 / ResourceNotFoundException